

Blood and its functions

Adults have about 5.5 litres of blood. The blood's function is to transport substances around the body.

About 55 per cent of blood is a straw-coloured liquid called **plasma**. The remaining 45 per cent consists of red and white cells, sometimes called corpuscles, and tiny particles called **platelets**.

Plasma

Plasma is made up of water and many important dissolved substances. Plasma transports dissolved food from the intestine to all parts of the body, and carbon dioxide from all parts of the body to the lungs. It carries urea from the liver to the kidneys where it is excreted, and distributes chemicals called hormones from the endocrine glands to various tissues. Plasma also contains the protein fibrinogen which plays an important part in the clotting of blood in wounds.

Red blood cells, or erythrocytes

Human red cells are bi-concave discs (Fig. 1). There are about 5 million in a cubic millimetre of blood. They have no nucleus and live for only four months when they are destroyed in the spleen and liver. Some of their chemicals, such as iron, are re-used in the bone marrow for the manufacture of new red cells. About two million red cells are destroyed and replaced every second in the human body.

These cells owe their red colour to **haemoglobin**, a chemical which enables them to transport oxygen from the lungs to all parts of the body.

White blood cells, or leucocytes

White cells are really colourless, and unlike red cells, they have a nucleus. Most types of white cell can change shape, like an amoeba.

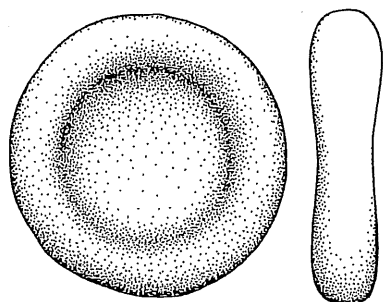
There are only 8000 white cells in a cubic millimetre of blood. About 75 per cent of white cells are made in the bone marrow; these cells have a large lobed nucleus, and are called **granulocytes** because they have tiny granules in their cytoplasm. The remaining 25 per cent of white cells are called **agranulocytes** because they have no granules. Most agranulocytes are made in the lymphatic tissues therefore they are sometimes called **lymphocytes** (Fig. 2).

The majority of leucocytes are **phagocytic**, a word which means 'cell eater'. They are described in this way because they destroy invading bacteria and dead cells by engulfing and digesting them. **Neutrophils** and **monocytes** are examples of phagocytic white cells (Fig. 2). These cells gather in wounds where they kill bacteria before they can enter the body. They are also capable of squeezing between the cells which make up blood capillary walls in order to reach bacteria anywhere in the body (Fig. 3).

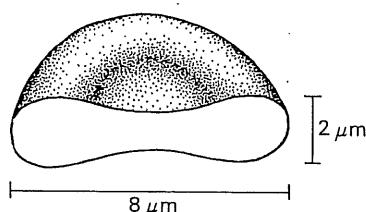
Platelets

Platelets are not complete cells. They are tiny fragments of cells made in the bone marrow. Their function during blood clotting is described in Unit 36.

A Front and side views of a red cell



B Red cell cut in half



C Red cells as they appear in a blood clot

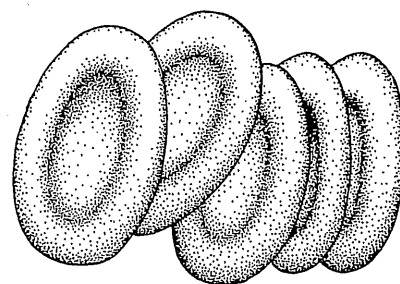


Fig. 1. Red blood cells have no nucleus and live for only four months, when they are destroyed in the spleen. New red cells are made in the bone marrow. Red cells contain haemoglobin, a

chemical which enables them to transport oxygen from the lung to all parts of the body.

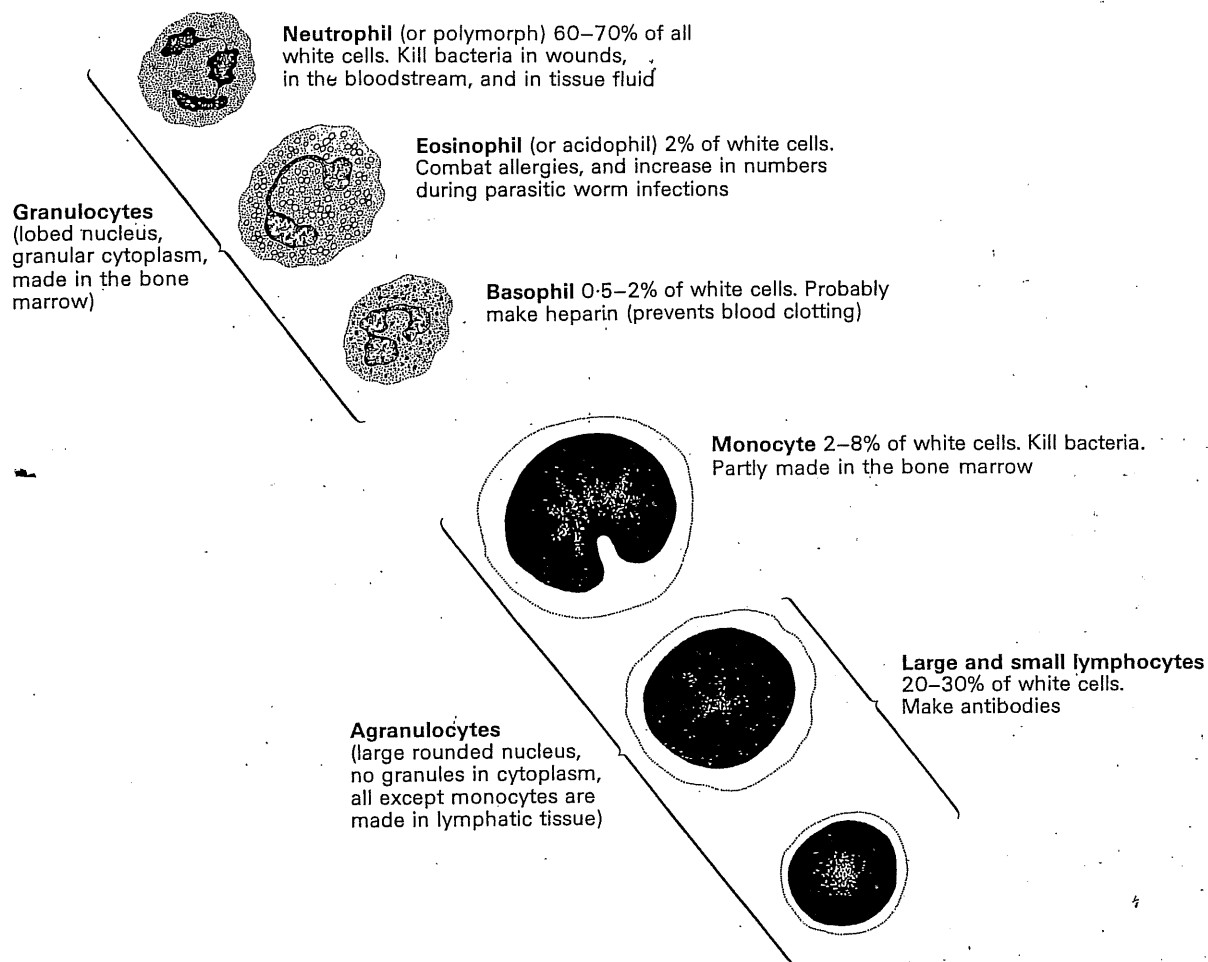


Fig. 2. The main types of white blood cell, or leucocyte, and their functions.

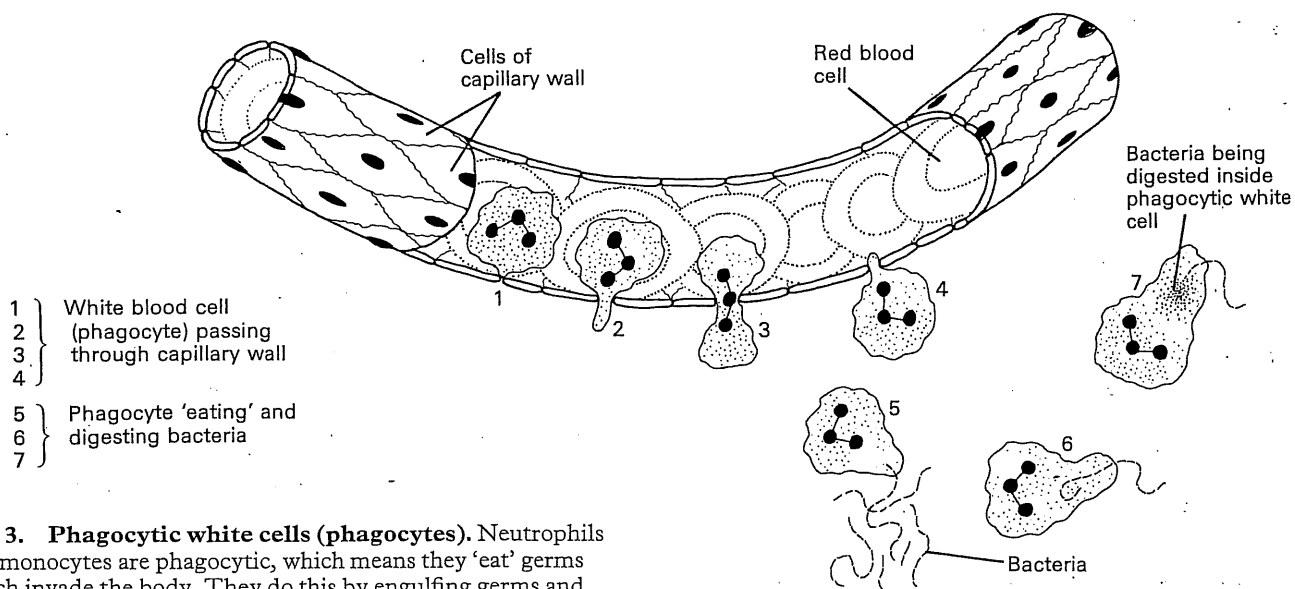


Fig. 3. Phagocytic white cells (phagocytes). Neutrophils and monocytes are phagocytic, which means they 'eat' germs which invade the body. They do this by engulfing germs and digesting them. Phagocytes can pass through capillary walls to reach germs in between body cells.

Structure of the heart

The heart is a hollow bag with walls made of **cardiac muscle**. This type of muscle is described in Unit 8. When the cardiac muscle contracts it squeezes blood out of the heart into vessels called **arteries** which carry it to all parts of the body. When the cardiac muscle relaxes it fills with blood from vessels called **veins**. These return blood from its journey around the body. One contraction and relaxation of the heart is called a **heart beat**. At rest the heart beats 60 to 70 times a minute. During this time it pumps about 5 litres of blood. Exercise may increase this rate to about 150 times a minute and then blood flow increases to 20 litres a minute.

The space inside the heart is divided into four

compartments called **chambers**. The two top chambers are called **atria**, or **auricles**, and their walls contain a thin layer of cardiac muscle. The two lower chambers are called **ventricles** and their walls are made of a much thicker layer of cardiac muscle.

The heart contains a number of **valves**. These control the direction in which blood flows through the heart. The **bicuspid** and **tricuspid** valves ensure that blood flows only from the atria into the ventricles. The lower edges of these valves are held in place by tendons and muscles. **Semi-lunar** valves are situated at the points where blood flows out of the heart. The semi-lunar valves ensure that blood cannot flow back into the ventricles.

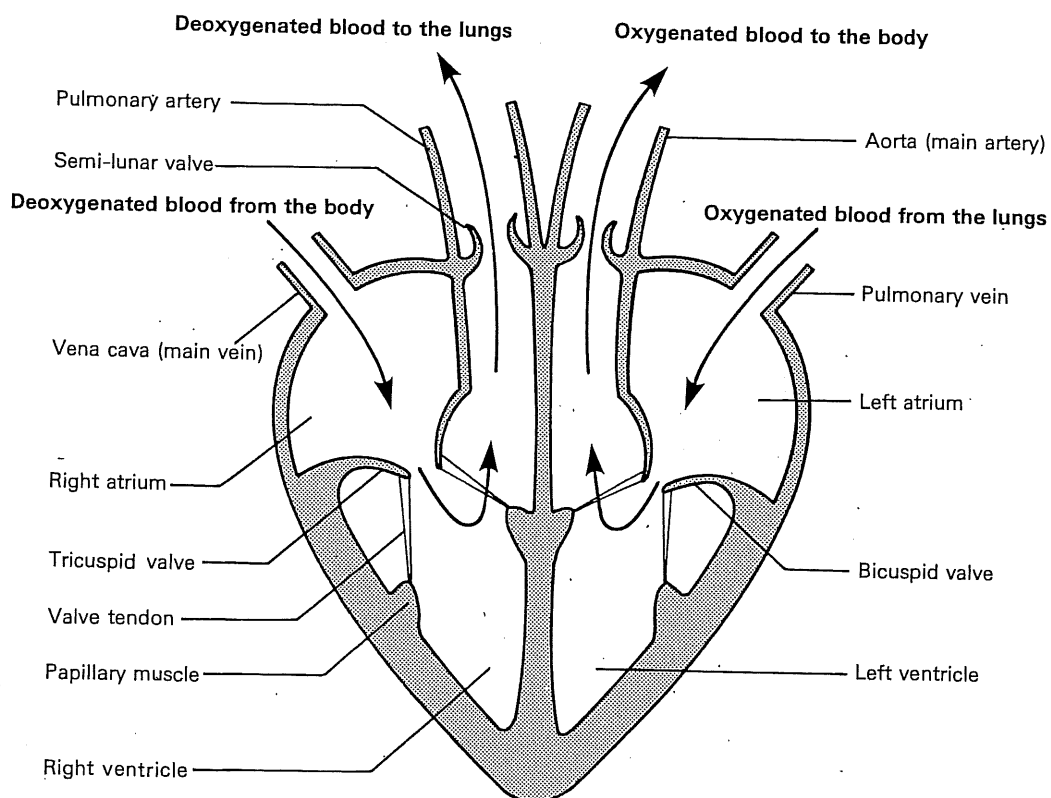


Fig. 1. Diagram of the heart showing the direction of the blood flowing through it. The heart is a bag whose walls are made of cardiac muscle. When this muscle contracts, blood is squeezed out of the heart and around the body. When the muscle relaxes, the heart fills with blood just returned from the body.

The heart has four compartments called chambers. Valves ensure that blood can only flow from the upper chambers—called atria—into the lower chambers—called ventricles. Valves at the exits from the heart ensure that blood cannot re-enter the heart by that route once it has left.

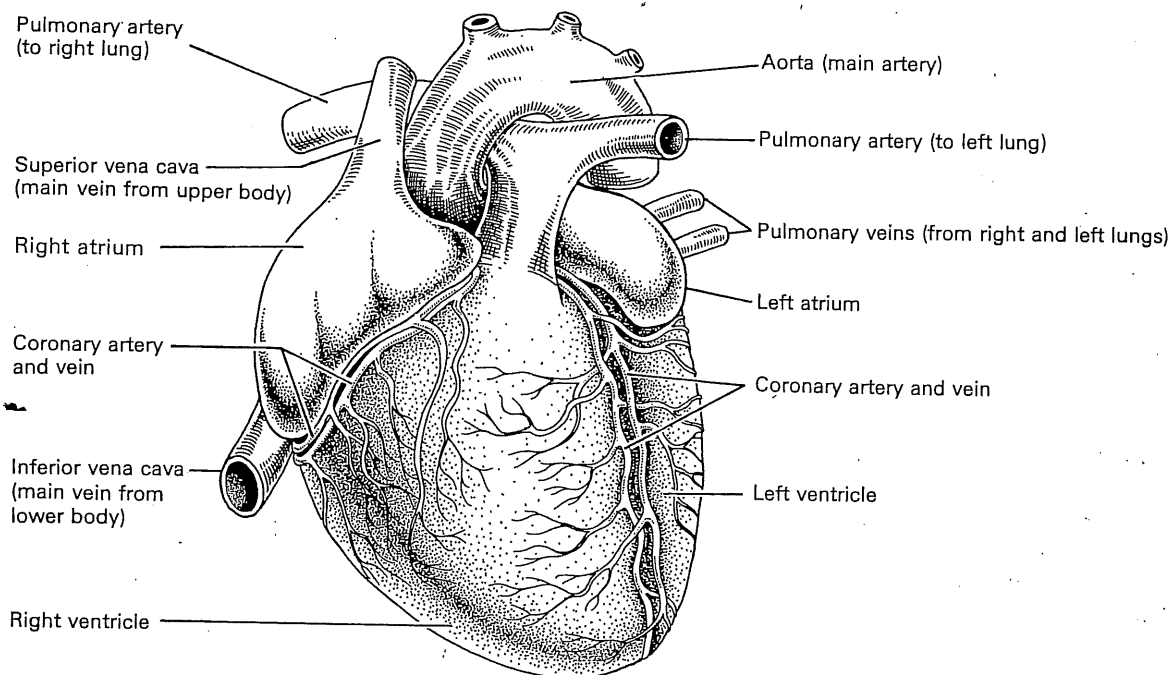


Fig. 2. **The human heart.** Note the coronary arteries and veins. These supply the heart's cardiac muscle with food and oxygen and remove its waste products.

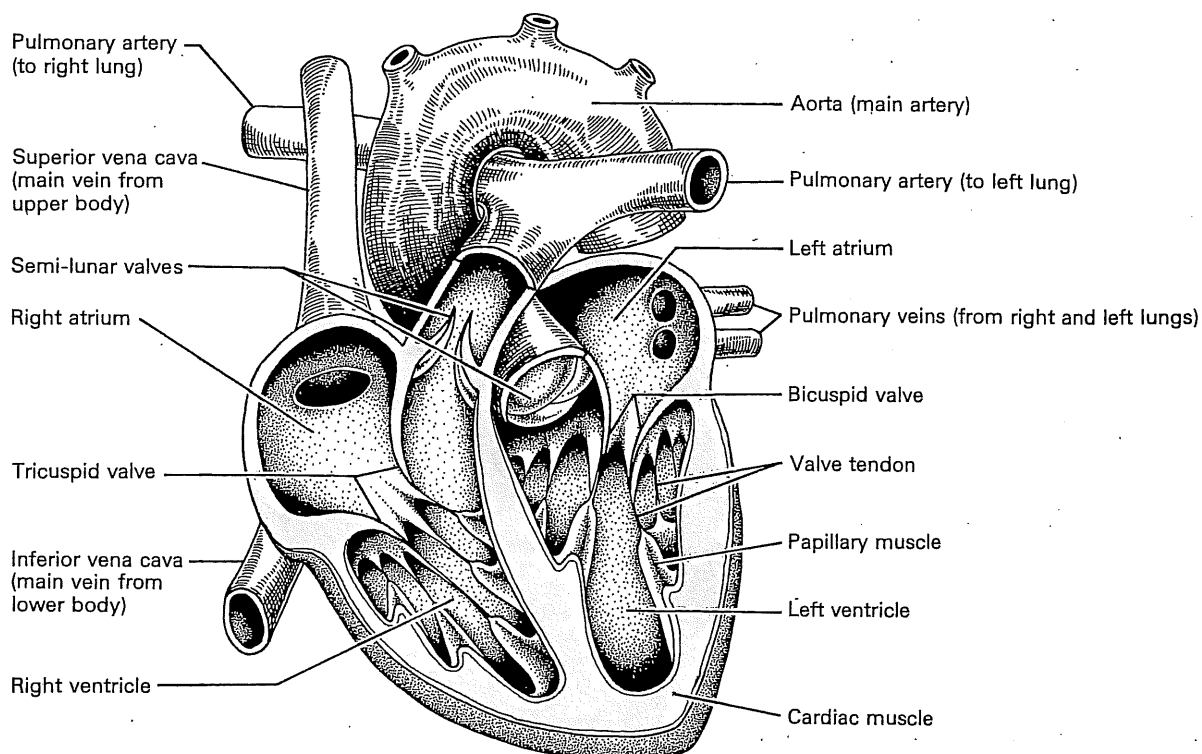


Fig. 3. **Drawing of a heart dissected to show its internal structure.**

Arteries, capillaries, and veins

The heart pumps blood into vessels called **arteries**. Arteries divide and sub-divide into narrower vessels called **arterioles** which, in turn, divide into extremely narrow vessels called **capillaries**. Eventually capillaries join together forming small veins called **venules**, and these lead into larger and larger veins which carry blood back to the heart.

Arteries and arterioles

Artery walls consist of three layers: an outer layer of fibrous tissue (elastic and collagen fibres); a middle layer of involuntary muscle and elastic fibres; and an inner layer only one cell thick called the **endothelium** which forms the lining of the vessel (Fig. 1A).

Walls of the larger arteries contain more elastic fibres than muscle. This allows them to expand and recoil with each heart beat as described in Unit 31. Small arteries and arterioles have more muscle than fibres in their walls. This muscle controls the diameter of the vessels making them narrower to reduce blood flow and wider to increase it.

Capillaries

Capillaries are the narrowest blood vessels (Fig. 1B). Some are so narrow that red blood cells are bent out of shape as they pass through. Capillary walls are a single layer of cells and are continuous with the lining of the larger vessels to which they are connected. Capillary walls are so thin that liquid from the blood is forced through them and then circulates between surrounding cells, carrying dissolved food and oxygen to them from the blood. This liquid is called **tissue fluid** and is described in Unit 34. Capillaries extend into almost every part of the body. None of the body's millions of cells is more than a fraction of a millimetre from one of these tiny vessels.

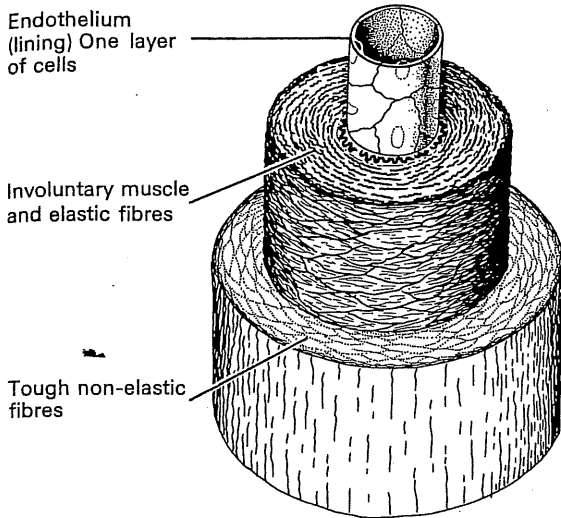
The amount of blood entering the capillaries is controlled by the diameter of the arterioles and by tiny rings of muscle where some arterioles join a capillary (Fig. 3). These rings of muscle are called **precapillary sphincters**. Flow of blood into a capillary network can be slowed down or stopped by these sphincters, and the blood diverted to other parts of the body.

Veins

Veins have thinner walls and wider passageways than arteries (Fig. 1C). Blood in veins is at a much lower pressure than in arteries, because pressure is lost as blood passes through the vast network of capillaries on its way to the veins. Flow of blood along veins to the heart is assisted in three ways. First, veins contain pocket valves which allow blood to flow only towards the heart. Second, many large veins are situated inside muscles of the legs, thighs, arms, etc. When these muscles are flexed the veins are squeezed and this squirts blood towards the heart. The valves prevent it flowing in the opposite direction (Fig. 2). If it were not for this mechanism, blood would gather in the feet and legs. Muscles are never still for long, even during sleep, and so blood flows along the veins at all times. Third, as the heart relaxes (diastole) its blood pressure is lower than in the veins and this causes blood to be sucked from the vein into the heart.

If the vein valves fail to close completely, blood gathers in the veins causing them to swell and become painful. This condition is called **varicose veins**. It is most likely to occur in the leg veins of people who have to spend long periods standing, or who frequently lift heavy weights. Such people should exercise their legs regularly, avoid standing still in one position for a long time, and rest with their feet up whenever possible.

Structure of an artery
The wall consists of three layers



B A capillary (not drawn to scale)
Wall consists of one layer of cells



C Structure of a vein
The two outer layers are thinner than those in an artery wall

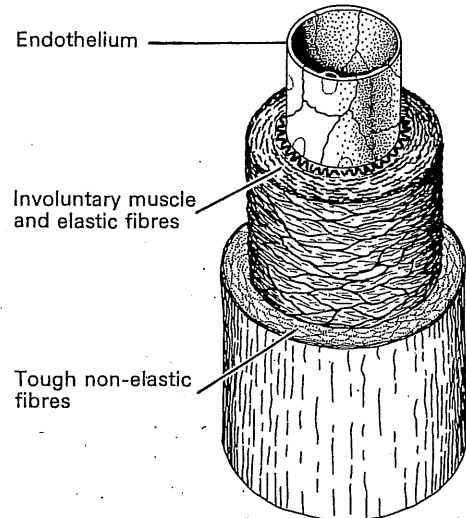


Fig. 1. Arteries, capillaries and veins. Arteries have thick walls and carry blood at high pressure away from the heart. Arteries divide to form extremely narrow vessels called capillaries. Capillary walls are so thin that liquid passes through

them from the bloodstream, carrying dissolved food and oxygen to surrounding cells. Capillaries join up to form veins. Veins carry blood at low pressure back to the heart and they contain valves which ensure that blood flows in only that direction.

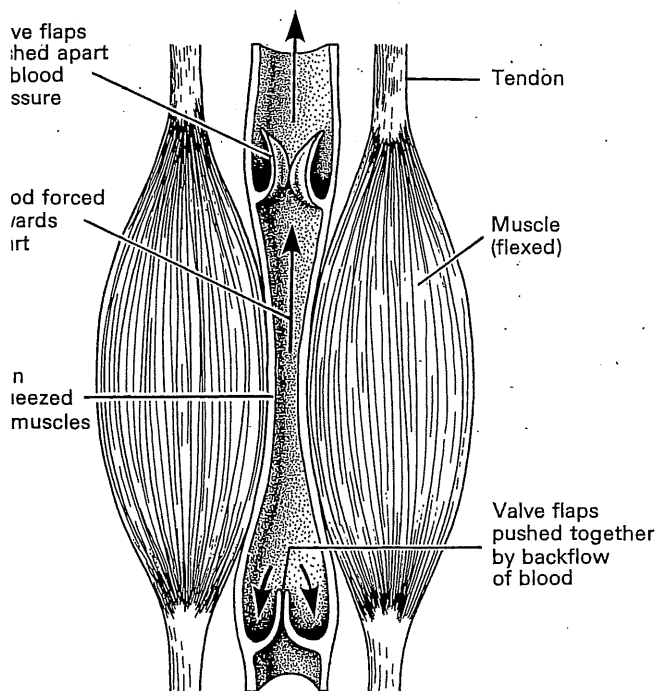


Fig. 2. Many large veins are situated between muscles. When the muscles are flexed the veins are squeezed, forcing blood in the direction which the valves allow—that is, towards the heart.

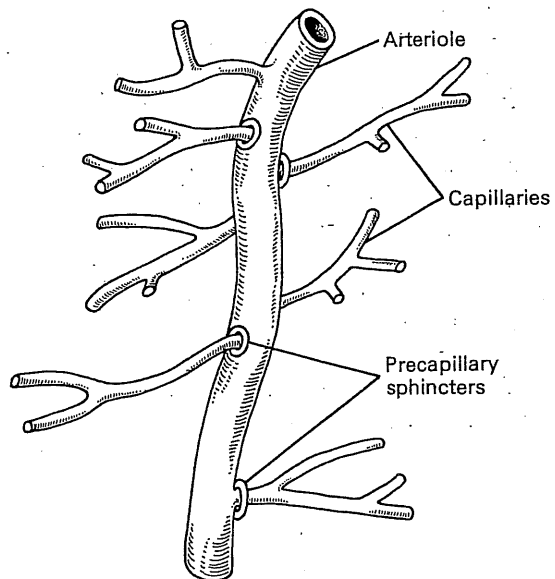


Fig. 3. Control of blood flow through capillaries. The amount of blood entering capillaries is controlled by the diameter of the arterioles and by tiny rings of muscle called precapillary sphincters where some capillaries branch off from arterioles. These sphincters can slow down or stop flow of blood into a capillary network, thus diverting blood to other parts of the body.

The main blood vessels

Humans and all other mammals have a *double* circulatory system: their blood circulates through two separate systems joined only at the heart. One system is called the **pulmonary circulation** and carries blood from the heart to the lungs and back. The other system is called the **systemic circulation** and carries blood from the heart to all parts of the body except the lungs and back again (Fig. 1).

Pulmonary circulation

Starting at the right ventricle, blood is pumped at high pressure along the pulmonary artery to the lungs. This blood is **deoxygenated**, which means it contains no oxygen, having released all of this gas on its journey around the body. The blood renews its supply of oxygen as it passes through the lungs; that is, it becomes **oxygenated** again. It then returns at

low pressure to the heart through the pulmonary vein.

Systemic circulation

Blood from the lungs enters the heart at the left atrium. It is pumped to the left ventricle and then pumped at very high pressure into the systemic circulation through the **aorta**, or main artery of the body. The aorta has branches leading to the head and arms, to the legs, and to organs in the abdomen (Fig. 2).

Two large veins return blood to the heart and complete the systemic circulation. The **superior vena cava** is the vein which carries blood from the head and arms, and the **inferior vena cava** returns blood from the legs and abdomen. Both of these veins empty into the right atrium.

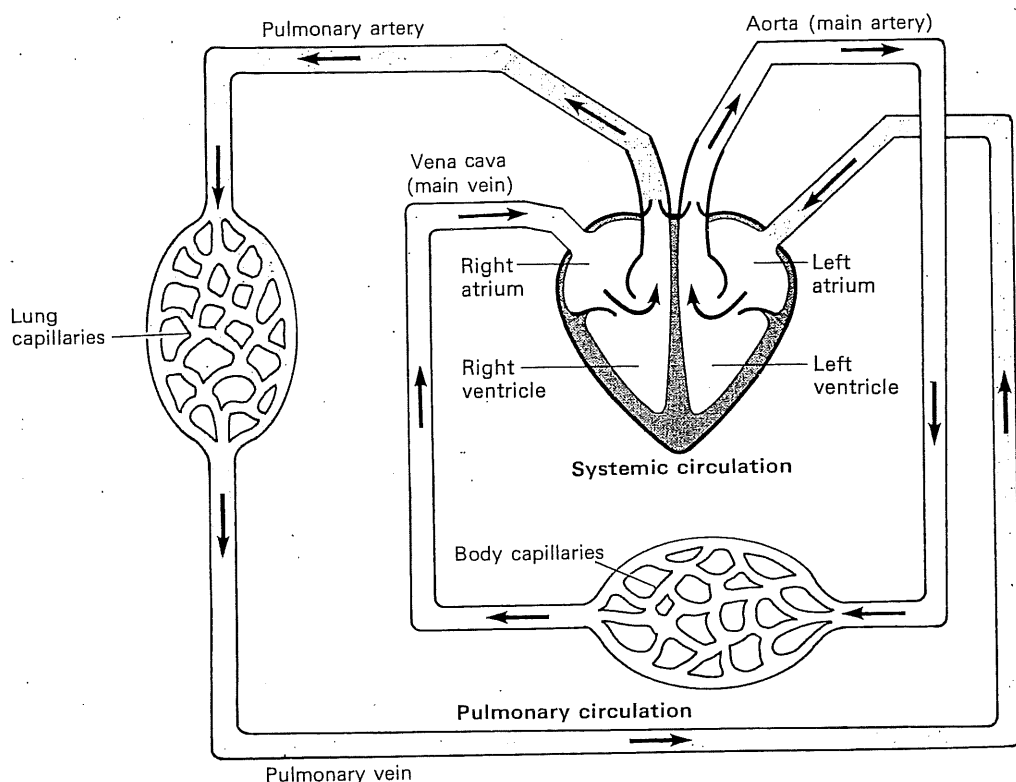
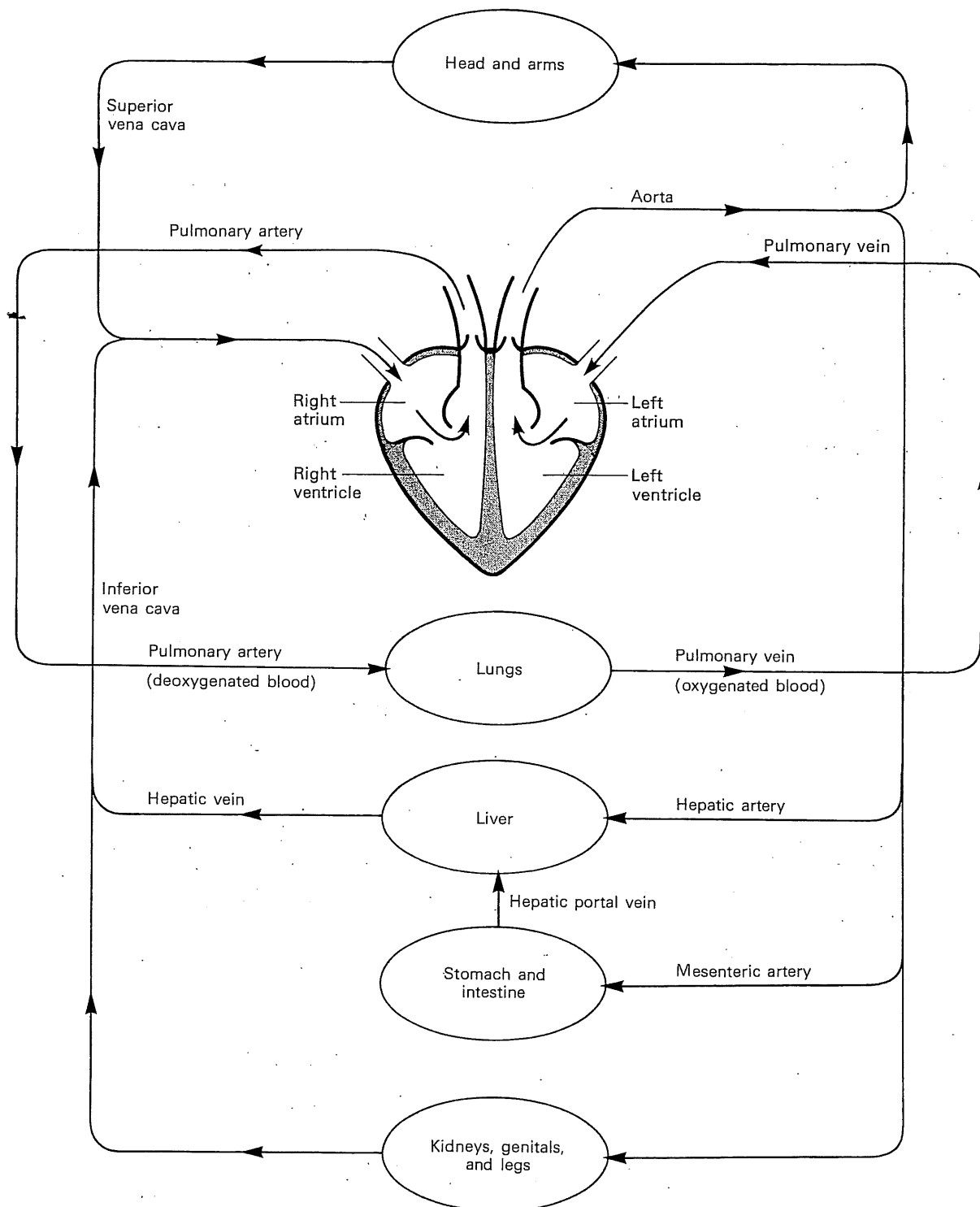
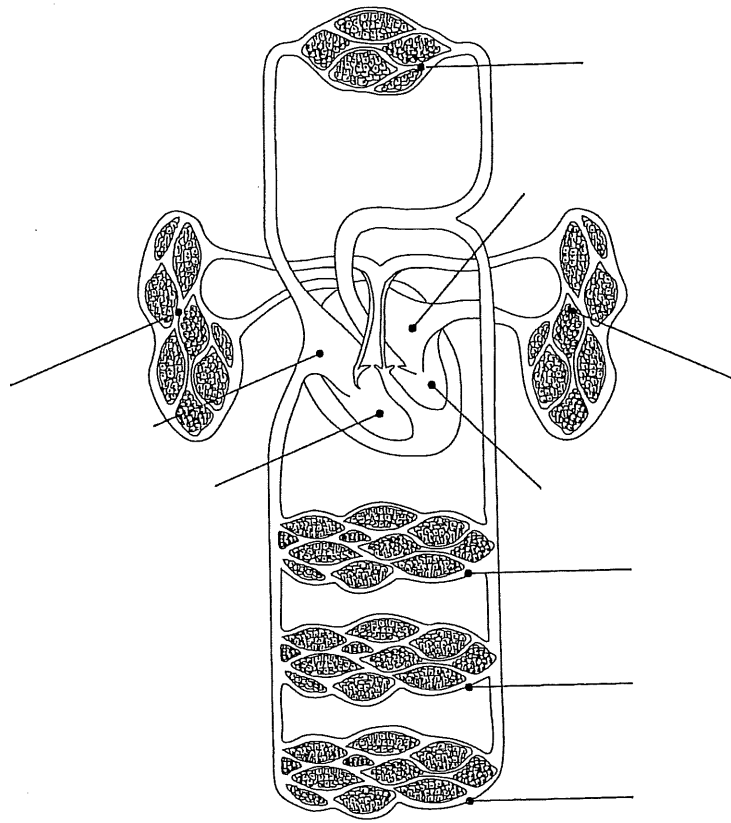


Fig. 1. Diagram of the double circulatory system of humans and other mammals. The pulmonary circulation carries blood from the heart to the lungs and back again. The systemic circulation carries blood from the heart to all parts of the body except the lungs and back again.



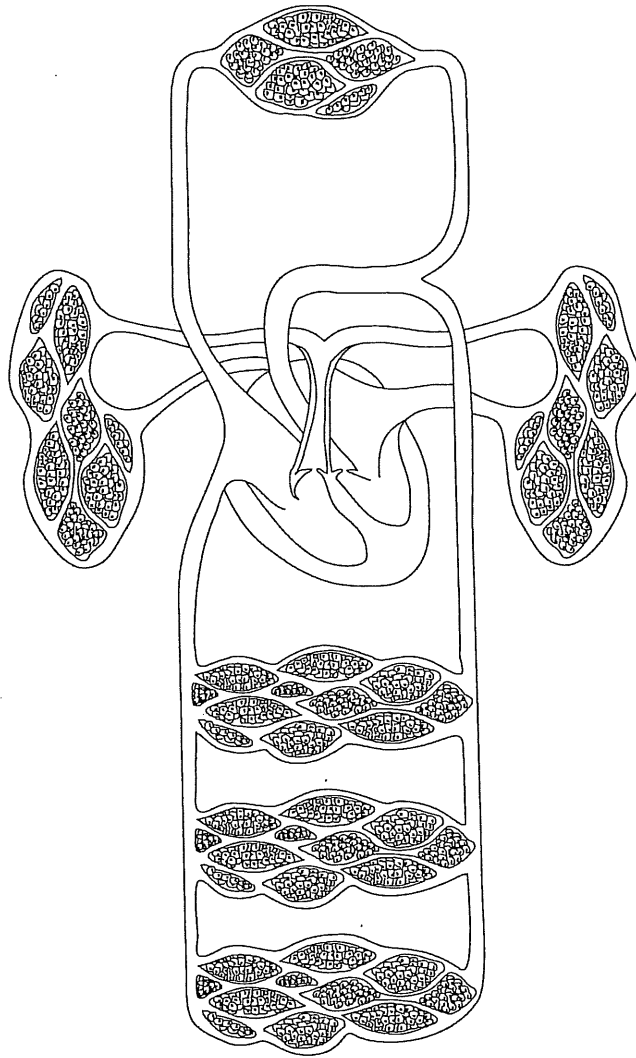
g. 2. Diagram of the main blood vessels.

6. Label the diagram below, which represents the body's circulatory system:



- (i) on the diagram:
- (a) using a pencil, draw arrows to indicate the direction in which the blood flows through this system
 - (b) using a red pen, shade in the arteries and veins which carry oxygenated blood
 - (c) using a blue pen, shade in the arteries and veins which carry deoxygenated blood
- (ii) In general terms, where does the blood become deoxygenated?
- _____
- (iii) Specifically where does the blood become oxygenated?
- _____
- (iv) Only one artery in the diagram carries deoxygenated blood. Which artery?
- _____
- (v) Only one vein in the diagram carries oxygenated blood. Which vein?
- _____
- (vi) What is carried by the hepatic portal vein, which makes it unique?
- _____

10. (i) On the diagram below, shade the pulmonary system (e.g. use blue) and using a different colour (e.g. use green) show the systemic circulation.

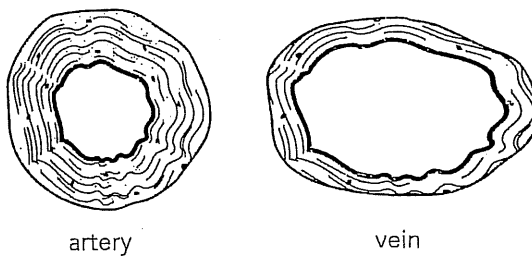


- (ii) Make a list of the arteries and veins that make up the:

Pulmonary circulation _____

Systemic circulation _____

11. The diagram below shows some differences between arteries and veins.



artery

vein

Complete the table below to describe the main differences.

FEATURE	ARTERY	VEIN
comparative thickness of wall		
comparative diameter of lumen		
presence of valves		
elasticity		

5.2 THE COMPONENTS OF BLOOD



Terminology

These are some of the terms from this section which you should know. Write the meaning of each term in the space provided.

(v) plasma

(vi) platelet (thrombocyte)

Review Questions

1. What type of tissue is blood and why is it classified in this category?

2. Describe five functions of blood.

- (i) _____
- (ii) _____
- (iii) _____
- (iv) _____
- (v) _____

3. (i) Make a list of the components of blood plasma.

(ii) What percentage of the blood is plasma? _____

(iii) The plasma is viscous. What does this mean?

4. How does carbon dioxide travel in the blood?

5. Describe two ways in which the blood is protective.

6. Forty-five per cent of the volume of blood is made up of formed elements (cells and cell fragments). The most important of these are illustrated in the table below. To complete the table name each, indicate its origin (where it is synthesised), what it does for the body and any other significant points about it.

DIAGRAM (not to scale)	NAME	ORIGIN	FUNCTION	OTHER NOTES
 side view				

Photocopying is prohibited by law without the prior written permission of the publisher. This book is a copyright work of the publisher. All rights reserved. No part of this book may be reproduced, stored in a retrieval system, or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or by any information storage and retrieval system, without the prior written permission of the publisher.

112

8. How does the circulatory system contribute to the efficient exchange of gases in the lungs?

9. (i) Red blood cells contain a protein which is important in transporting oxygen around the body. What is the name of this protein?

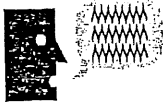
- (ii) Red blood cells have no nucleus. What advantage does this give?

- (iii) What is the disadvantage of having no nucleus?

- (iv) Where are old dysfunctional red blood cells broken down?

- (v) Name the useful product formed from this breakdown process.

5.3 THE LYMPHATIC SYSTEM



Terminology

These are some of the terms from this section which you should know. Write the meaning of each term in the space provided.

- (i) lacteal

- (ii) lymph

- (iii) lymph node

- (iv) lymph valve

- (v) lymphocyte

Refer to your textbook pages 100-102 to complete the following summary of the Circulatory System.

Purpose and Function of the Circulatory System (Fill in the blanks using the terms below.)

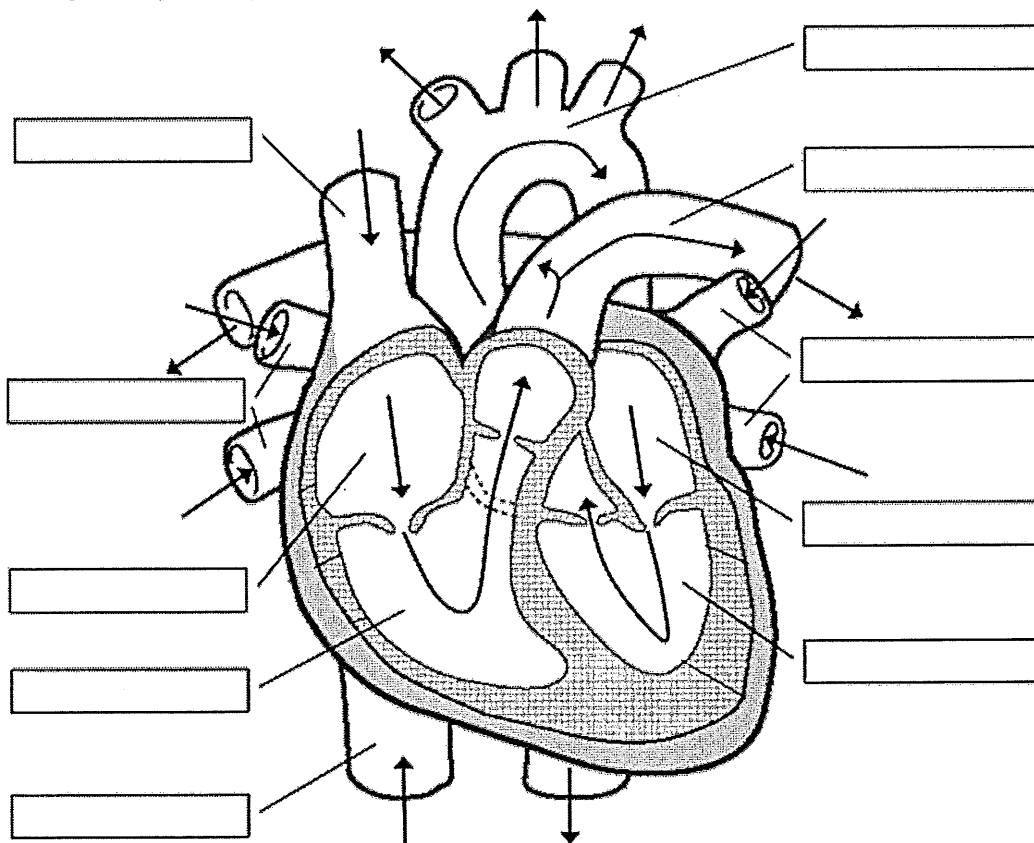
valves exchange arteries veins contracts capillaries oxygen wastes nutrients

The function of the circulatory system is to transport _____ and _____ to cells and carry _____ away from cells to the organs responsible for eliminating them from the body.

The heart is a muscular pump. When the heart _____, it produces pressure on the blood in the circulatory system. This pressure pushes blood through the body. The body travels away from the heart in vessels called _____. It returns to the heart through vessels called _____. To stop blood from flowing backwards, flexible flaps of tissue called _____ are found throughout the circulatory system in the heart and veins. Arteries and veins are connected by tiny, thin-walled vessels called _____. These very small vessels are in close contact with the organs and tissues of the body. The _____ of nutrients, wastes, oxygen and carbon dioxide occur between tissues and the blood through the walls of these small vessels.

Label the diagram using the bold words in the paragraph below, then colour it red where oxygenated blood is and blue where deoxygenated blood is. Use the paragraph and the picture on page 69 of your textbook to help you.

The heart is a fist-sized, muscular organ that pumps blood through the body. Deoxygenated blood enters the **right atrium** of the heart via veins called the **inferior vena cava** and the **superior vena cava** (superior means "above" and inferior means "below"). The blood is then pumped to the **right ventricle** and then through the **pulmonary artery** to the lungs, where it is enriched with oxygen and loses carbon dioxide. The oxygenated blood is then carried back to the **left atrium** of the heart via the **pulmonary veins** (label twice). The blood is pumped to the **left ventricle**, then through the **aorta** to the rest of the body. This cycle is repeated.



Now, it is time to see how much you have learned!
Take the quiz below on the blood vessels found in your body!

Quiz on Blood Vessels

Good Luck!

Fill-in the Correct Answer Codes

1 Which is the largest vein in the human body?

A

Capillaries

2 Which type of blood vessels have thin non-elastic walls & carry blood back to the heart?

B

Pulmonary

3 All arteries carry oxygen-rich blood except for which artery?

C

Veins

4 Which type of blood vessels contain valves to prevent the back-flow of blood?

D

Arteries

5 Which type of blood vessels have thick elastic walls & carry blood away from the heart?

E

Aorta

6 Which is the largest artery in the human body?

F

Inferior vena cava

7 Which blood vessels allow exchange of gases and nutrients between blood and the body cells?

G

Capillaries

8 Name the tiny, thin-walled blood vessels that connect arteries and veins?

H

Veins

3.6 The heart

Science inquiry skills

FOUNDATION

STANDARD

ADVANCED

Processing
& Analysing

Communicating

- 1 A diagram of the heart (Figure 3.6.1) is provided below.
 - (a) Add labels from the box to identify the parts of the heart.
 - (b) Lightly colour the heart and blood vessels to identify where there is oxygenated blood (red) and deoxygenated blood (blue).
 - (c) Add arrows to identify the direction of blood flow through the heart.
 - (d) At the end of the blood vessels, name the part of the body the blood is flowing to or coming from.

aorta
left atrium
left ventricle
pulmonary artery
pulmonary vein
right atrium
right ventricle
valves
vena cava

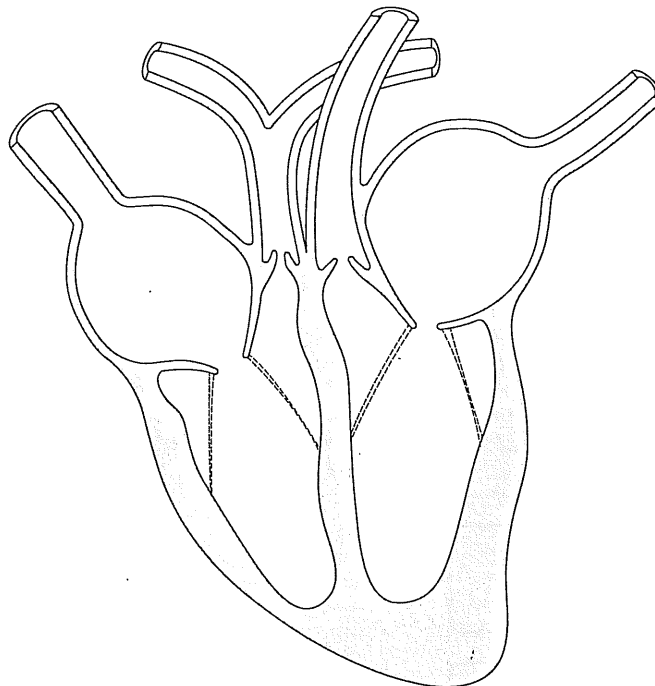


Figure 3.6.1 The human heart

- 2 (a) Identify whether the right or left ventricle is larger and has thicker walls.

- (b) Explain why this ventricle has thicker walls.

- 3 Construct a flow diagram to show the passage of the blood through the body and heart. Start and end with the right ventricle.

RATE MY UNDERSTANDING
Shade the face that shows your rating

